Zero-inflated models, dependent data

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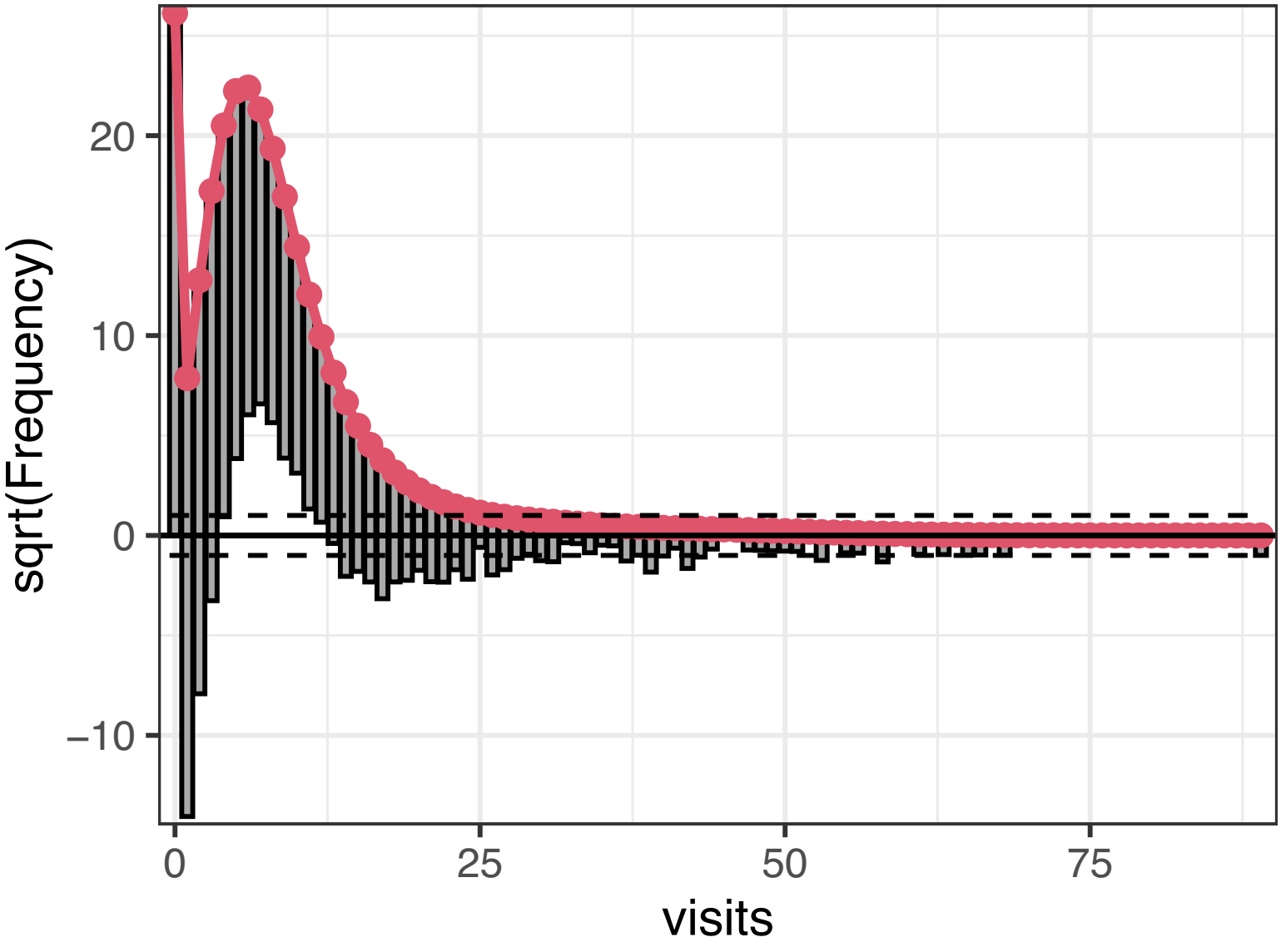
Physician visits

Data on 4406 individuals, age 66 and over, who are covered by Medicare. Variables include:

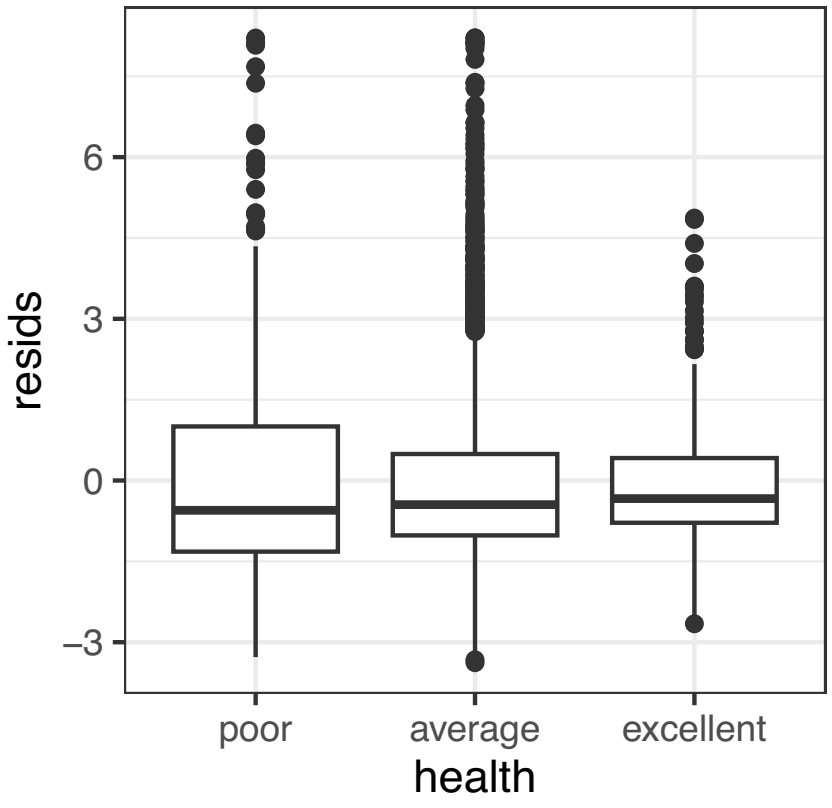
- `visits`: Number of physician office visits
- `hospital`: Number of hospital stays
- `health`: Factor indicating self-perceived health status, levels are "poor", "average", "excellent"
- `chronic`: Number of chronic conditions
- `gender`: Factor indicating gender
- `school`: Number of years of education
- `insurance`: Factor. Is the individual covered by private insurance?

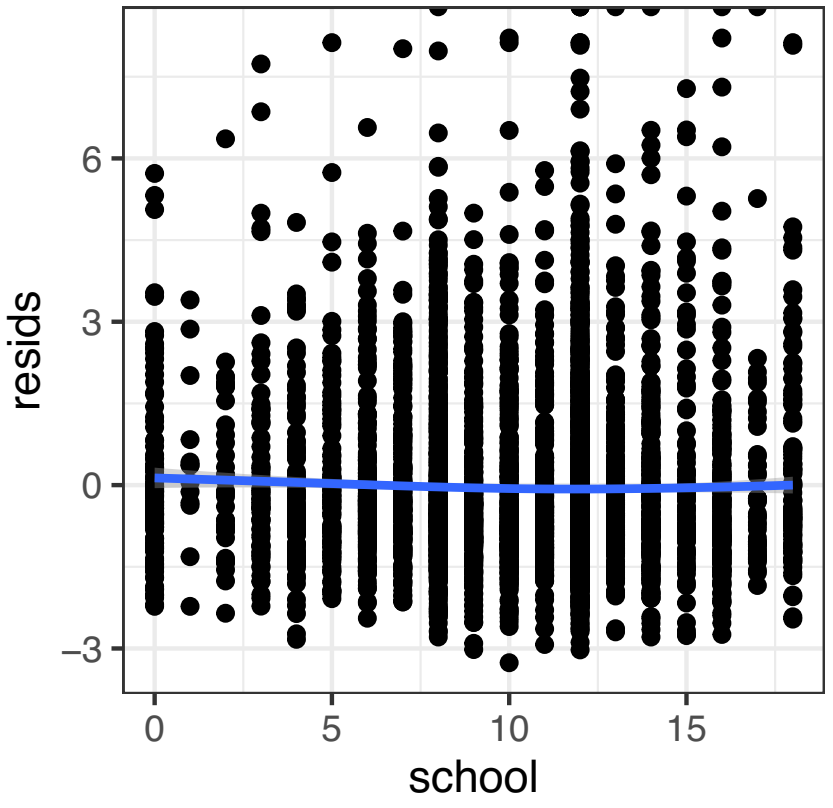
Goal: Model the number of physician visits.

ZIP diagnostics

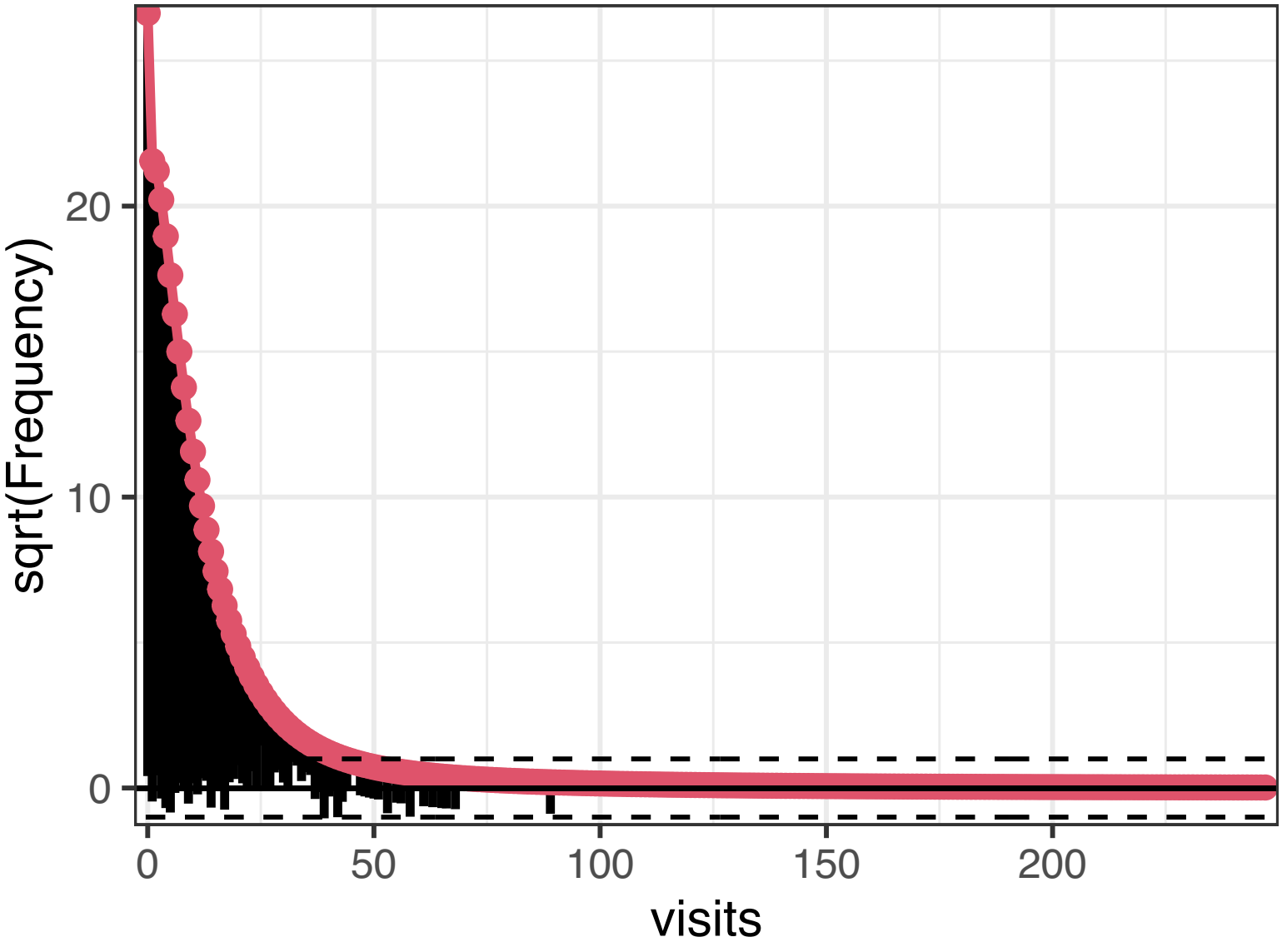


ZIP diagnostics

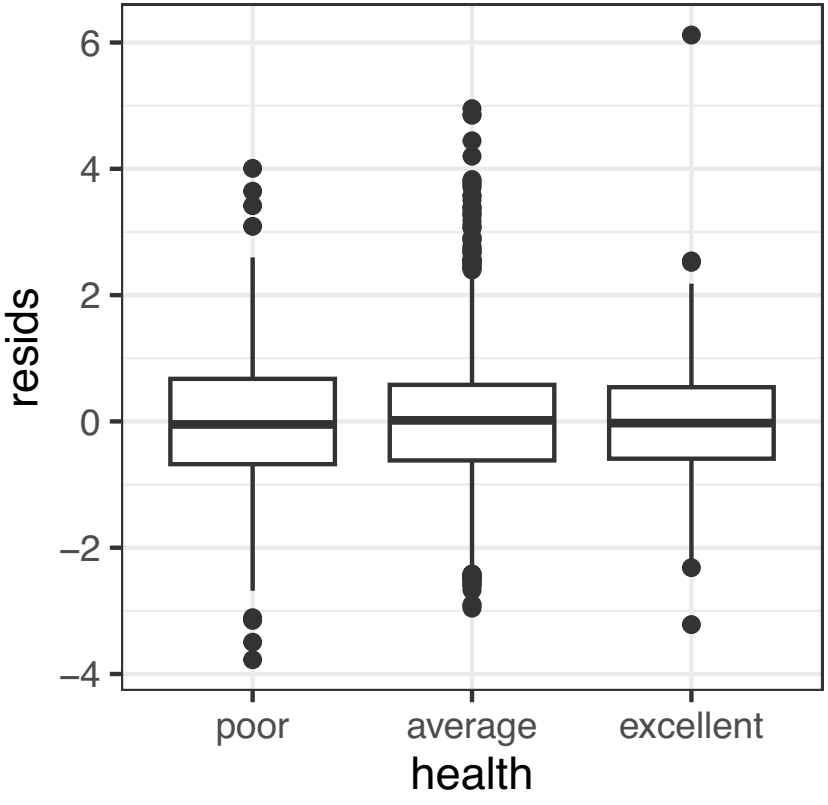


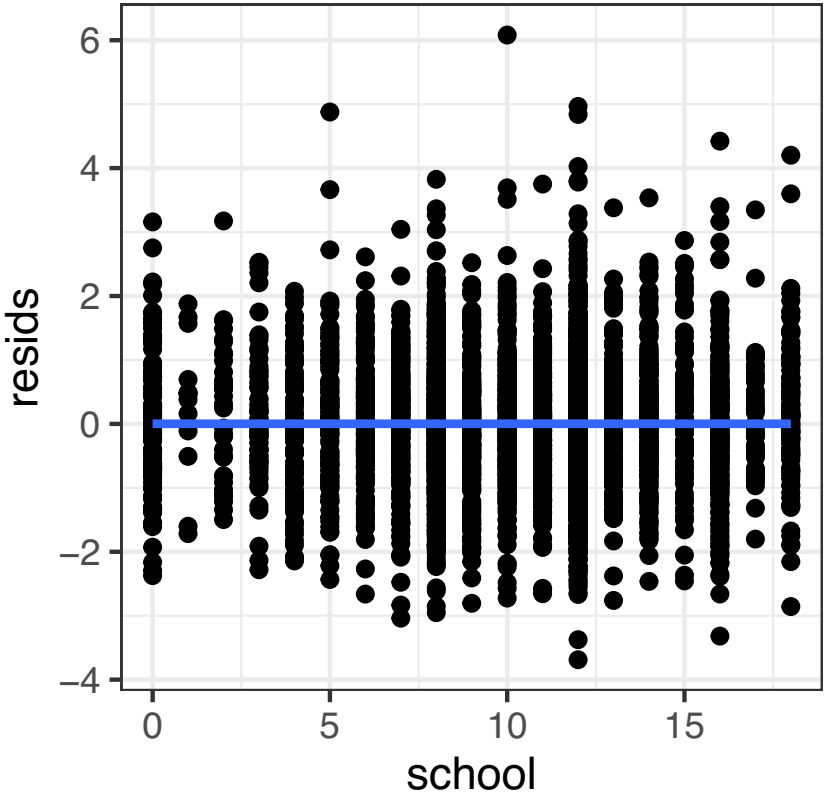


ZINB diagnostics



ZINB diagnostics





Inference

We are interested in whether there is any relationship between an individual's self-perceived health status (`health`) and their number of physician visits (`visits`), after accounting for other explanatory variables.

Question: How would I carry out a hypothesis test to address this question?

Likelihood ratio test:

reduced model drop terms for health

from both Poisson & logistic pieces of model

health is categorical w/ 3 levels

Inference

```
m2 <- pscl::zeroinfl(visits ~ .,
                     dist = "negbin", data = nmes)
m3 <- pscl::zeroinfl(visits ~ . - health,
                     dist = "negbin", data = nmes)

2*(m2$loglik - m3$loglik)
```

```
## [1] 70.85807
```

```
pchisq(2*(m2$loglik - m3$loglik), df=4, lower.tail=F)
```

```
## [1] 1.495596e-14
```

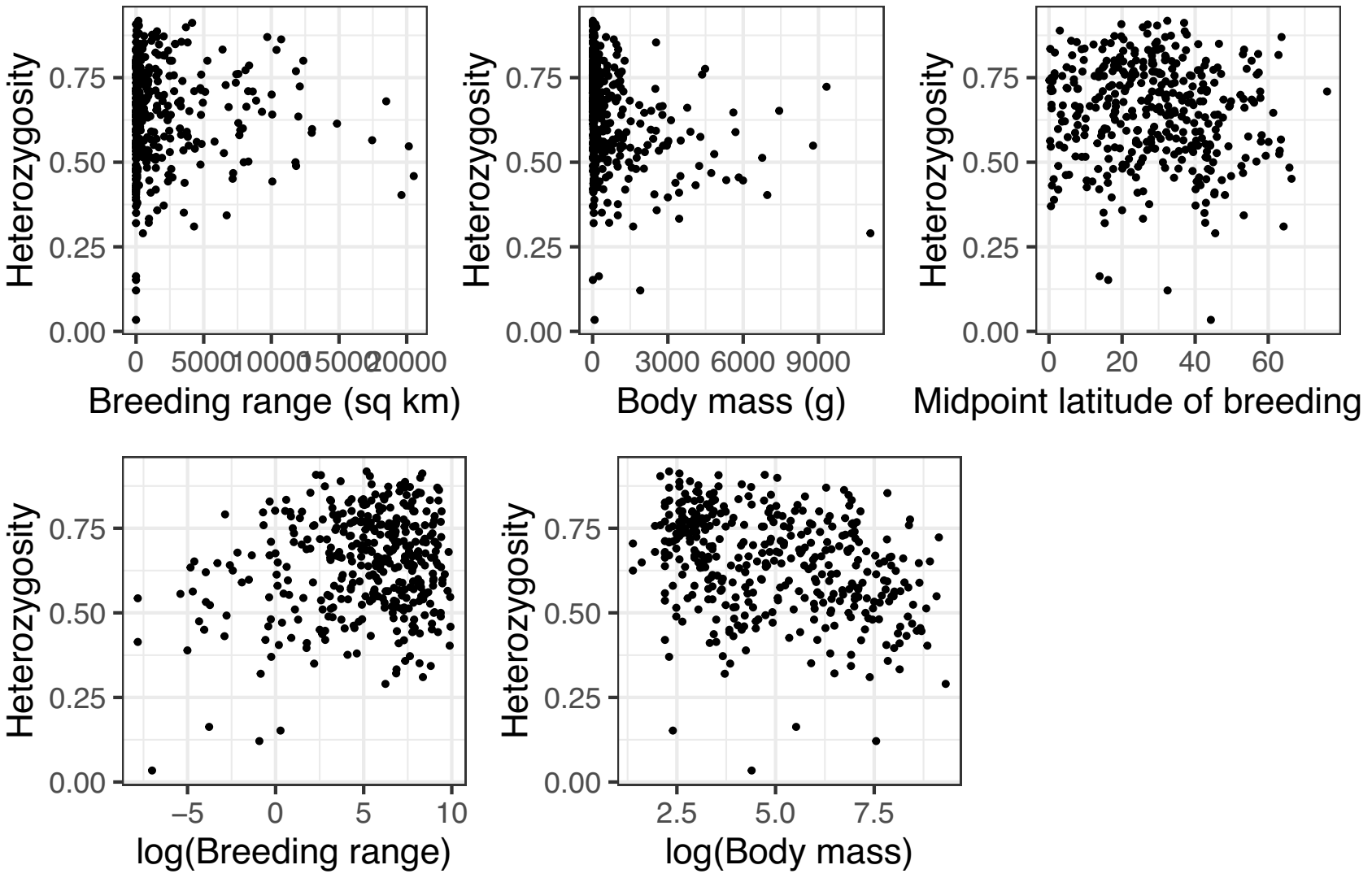
Another dataset

Data on 387 different bird species, with information from each species that includes:

- `Species`: the name of the species
- `Family`: the family of birds to which the species belongs (remember that in taxonomy, "family" is more general than "species" – each family consists of a set of related species)
- `Heterozygosity`: a measure of species genetic diversity. Each bird has two copies of each chromosome, and heterozygosity measures the probability that the two chromosomes will have different versions (alleles) for a genomic marker
- `breeding_range`: the breeding range of the species (in 10000 sq. km)
- `body_mass`: species average body mass (g)
- `Latitude`: mid-point latitude of species breeding range

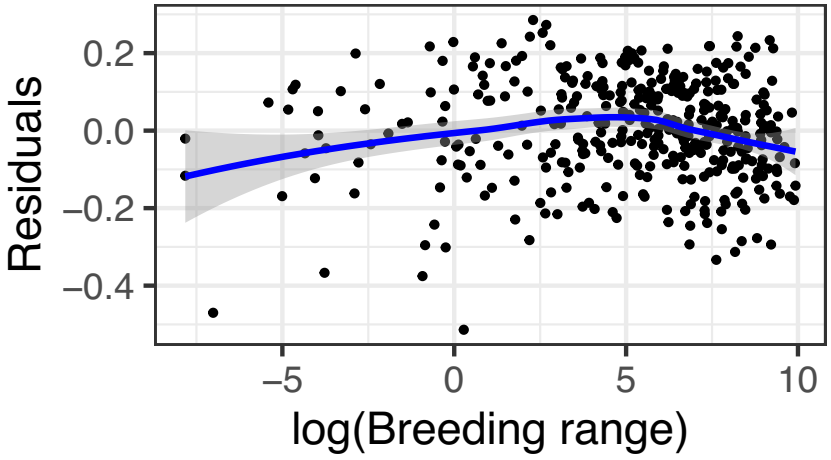
Goal: Model relationship between breeding range and species diversity.

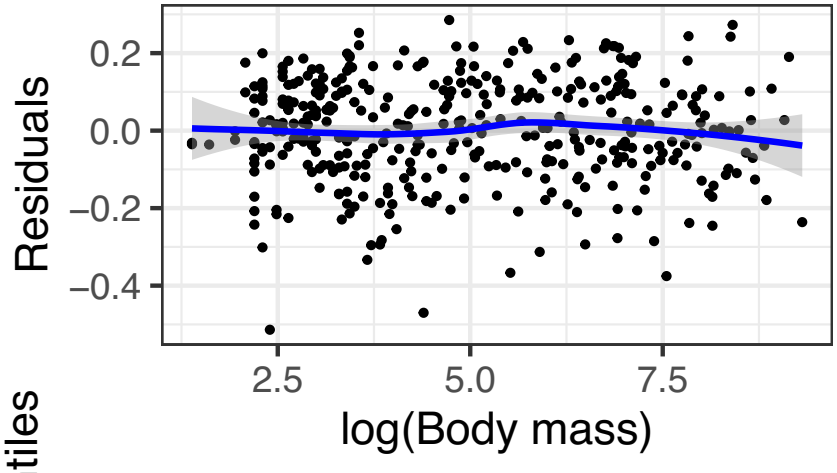
EDA

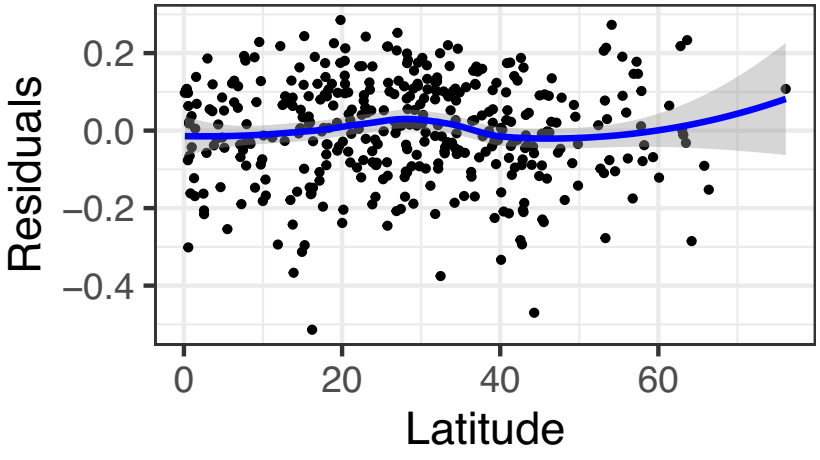

Heterozygosity
0.00
0.25
0.50
0.75
0
5000
10000
15000
20000
Breeding range (sq km)
0
3000
6000
9000
Body mass (g)
0
20
40
60
Midpoint latitude of breeding
−5
0
5
10
log(Breeding range)
2.5
5.0
7.5
log(Body mass)

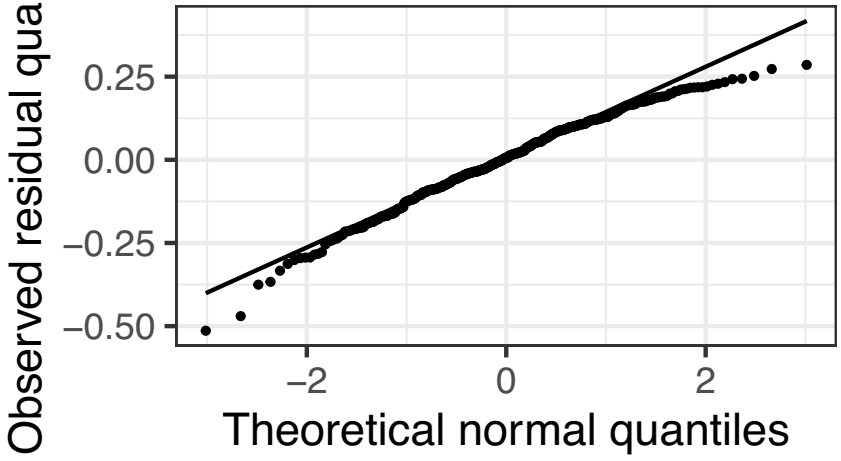
Linear regression

```
bird_lm <- lm(Heterozygosity ~ log(breeding_range) +
                log(body_mass) + Latitude,
              data = bird_diversity)
```



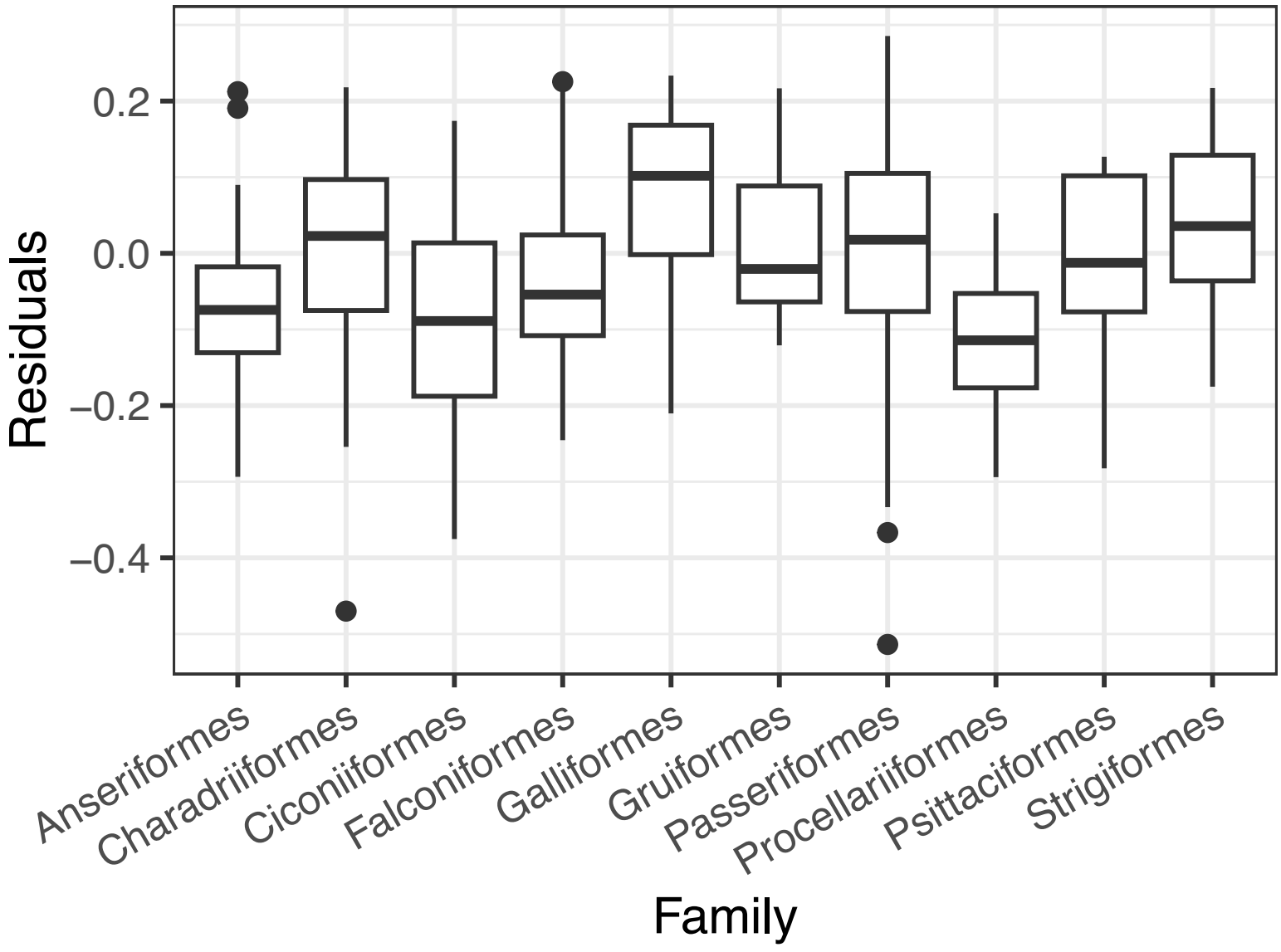






Family?

Residuals for the 10 largest families:



Family?

Current model:

$$\text{Heterozygosity}_i = \beta_0 + \beta_1 \log(\text{breeding range}_i) + \beta_2 \log(\text{mass}_i) + \beta_3 \text{latitude}_i + \varepsilon_i$$

Question: How many parameters would be added to the model if I want to include Family as an explanatory variable?

Family

Current model:

$$\text{Heterozygosity}_i = \beta_0 + \beta_1 \log(\text{breeding range}_i) + \beta_2 \log(\text{mass}_i) + \beta_3 \text{latitude}_i + \varepsilon_i$$

Adding Family:

$$\text{Heterozygosity} = \beta_0 + \sum_{i=1}^{19} \beta_i \mathbb{I}\{\text{Family} = i\} + \beta_{20} \log(\text{breeding range}) + \cdots + \varepsilon$$

Challenges

1. There are 20 different families, so including Family as a variable would require 19 additional β coefficients (or more, if any interactions are included!).

2. There are three families with only a single species in the dataset, and one family with only two species. Estimating a coefficient using only one or two points is not a good idea.

3. While we want to capture differences between families in the model, we probably don't care about the actual coefficient estimates, because our main variable of interest is breeding range. Rather, we just want our model to capture variability from family to family.

Alternative: linear mixed model

$$\text{Heterozygosity}_{ij} = \beta_0 + u_i + \beta_1 \log(\text{breeding range}_{ij}) + \beta_2 \log(\text{mass}_{ij}) + \beta_3 \text{latitude}_{ij} + \varepsilon_{ij}$$

$u_i \overset{iid}{\sim} N(0, \sigma_u^2)$

$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$

i = family $\quad i = 1, \ldots, 20$

j = species w/in family $\quad j = 1, \ldots, n_i$ $\quad (n_i = \#\text{ obs in group } i)$

e.g. n_i ranges from 1 to 100+

u_i = random value for family i (random effect)

(capture differences between families)

$\beta_0 + u_i$ = intercept for family i (random intercept)

ε_{ij} captures random variability between species within families

$\beta_1, \beta_2, \beta_3$: fixed effects (fixed but unknown parameters, not random variables)